Name: Debayan Kundu
Phone No.: 9775343416
Univ Roll No.: 13000124055
Date of Birth: 07/10/2005

Operating system – WCA
Assignment (PCC-CS-502)

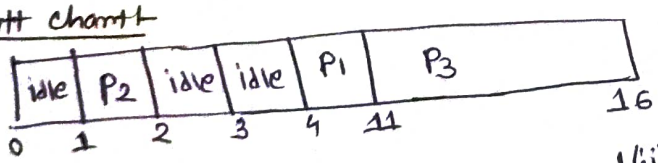
1) consider a uniprocessor system, where three process (process ID 1,3,3) are there. Arrival time of the three processes are last 3 digit of your mobile number (i.e. if your mobile number is 9876540123, arrival time of P1=1, P2=2, P3=3). Burst time process 1 is the addition of date (dd, i.e, 21/07/2003 is DOB, then it is 2+1=3), Process 2 is addition of month (mm, if July then 07 i.e. 7) and process 3 is addition of last two digit of the year (yy if then if 2003 then 0+3 i.e. 3 if it is 2000 first 2 digit). Assume CPU is using SRTF algorithm-

(i) calculate waiting time of each process.
(ii) calculate turnaround time of each process.
(iii) calculate percentage of time CPU remains idle.

⇒ Lats 3 digit of mobile number = 416
P1=4, P2=1, P3=6
DOB= 07/10/2005.
Burst time: P1= 0+7=7, P2= 1+0=1, P3=0+5=5

	Burst time	Arrival time
1. P1	7	4
2. P2	1	1
3. P3	5	6

Gantt chart:



TAT = Turn around time = completion time – arrival time
Waiting time = TAT – Burst time

(i) P1: waiting time = 7−7=0
P2: waiting time = 1−1=0
P3: waiting time = 10−5=5

(ii) P1: turn around time = 11−4=7
P2: " " = 2−1=1
P3: " " = 16−6=10

(iii) % of time CPU remains idle = $\frac{(1-0)+(3-2)+(4-3)}{16} \times 100$ %

= 18.75 %

2) Consider a uniprocessor system, where there exist three processor (Process Id 0,1,2). Consider a Round Robin scheduling algorithm where time quantum is 2 unit time. Each of the process will arrive based on the last 3 digits of your roll number. (If Roll No. is: 13000124055, arrival time P0 = 0, P1 = 5, P2 = 5). CPU Burst time will be first three digit of your mobile number is 8756540939, CPU burst time of P0 = 9, P1 = 7, P2 = 5, if any number is 0, then take next non zero value from your mobile number like 7000312867, CPU Burst time of P0 = 7, P1 = 3, P2 = 1).

(i) Calculate Average waiting time.
(ii) Calculate Average Turnaround time.
(iii) Calculate the percentage of time CPU remains idle

⇒ Roll: 13000124055

Arrival time
P0 = 0, P1 = 5, P2 = 5

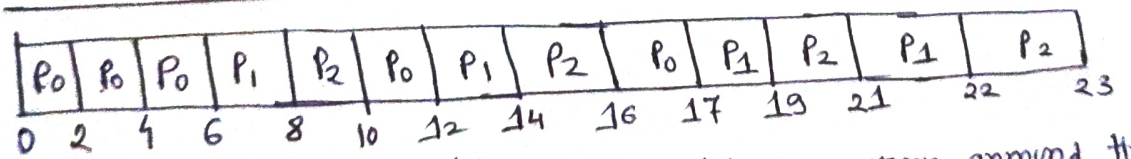
Mobile number: 9775343416

Burst time
P0 = 9, P1 = 7, P2 = 7

	Burst time	Arrival time
1. P0	9	0
2. P1	7	5
3. P2	7	5

time quantum = 2

Gantt chart



(i) P0: waiting time = 17 − 9 = 8
P1: " " = 17 − 7 = 10
P2: " " = 18 − 7 = 11

Avg. waiting time = $\frac{8+10+11}{3}$ = 9.66

(ii) P0: Turn around time = 17 − 0 = 17
P1: " " = 22 − 5 = 17
P2: " " = 23 − 5 = 18

Avg. turn around time = $\frac{17+17+18}{3}$ = 17.33

(iii) Idle time = 0
∴ percentage time CPU remains idle = $\frac{0}{23} \times 100\%$
= 0 %

3) Find out the no. of page faults will occur for optimal and LRU page replacement algo having 3 frames. Reference string will be the alternate number of your university Roll no. and mobile number (If your university roll number is 13000120012 and mobile number is 9831156378, then your string will be - 1,9,3,8,0,3,0,1,0,1,1,5,2, 6,0,3,0,7,1,8,2).

⇒ Roll = 13000124055 mobile numbe = 9775343416

String :-

1,9,3,7,0,7,0,5,0,3,1,4,2,3,4,4,0,1,5,6,5.

Optimal page replacement

1	9	3	7	0	7	0	5	0	3	1	4	2	3	4	4	0	1	5	6	5
		3	3	3	3	3	3	3	3	3	3	3	3	3	3	0	0	0	6	6
	9	9	7	7	7	7	5	5	5	1	4	4	4	4	4	4	1	1	1	1
1	1	1	1	0	0	0	0	0	0	0	0	2	2	2	2	2	2	5	5	5
*	*	*	*	*	H	H	*	H	H	*	*	*	H	H	H	*	*	*	*	H

* = Miss
H = Hit

Page fault = 13
Page Hit = 8

Least Recently used (LRU):-

1	9	3	7	0	7	0	5	0	3	1	4	2	3	4	4	0	1	5	6	5
		3	3	3	3	3	5	5	5	1	1	1	3	3	3	3	1	1	1	1
	9	9	9	0	0	0	0	0	0	0	4	4	4	4	4	4	4	5	5	5
1	1	1	7	7	7	7	7	7	3	3	3	2	2	2	2	0	0	0	6	6
*	*	*	*	*	H	H	*	H	*	*	*	*	*	H	H	*	*	*	*	H

Hit = 6
Page fault = 15